

ST. XAVIER'S COLLEGE

MAITIGHAR, KATHMANDU, NEPAL

Phone: 01-5321365, 01-5344636

Email: ktm@sxc.edu.np



LAB/WRITTEN ASSIGNMENT NUMBER: 4.2

“Spreadsheet Software”

Submitted By	Submitted To	Signature
Name: Abhishek Acharya Roll No: 721 Class: XI Section: G1	Department of Computer Science, NEB (Plus Two) St. Xavier's College	

Submission Date: 4th August 2026

Table of Contents

Introduction to MS Excel.....	1
Cell in MS Excel.....	3
Relative Cell Reference	3
Mixed Cell Reference.....	4
Absolute Cell Reference.....	4
VLOOKUP.....	5
Syntax of VLOOKUP.....	5
Marksheets.....	8-12
Formulas.....	17
Reflection.....	18
Conclusion.....	18

List of Figures

Figure 1: Structural Blueprint of the Microsoft Word User Interface	2
Figure 2: Screenshots of Sheet 1: Student Mark sheet Calculation	6
Figure 3: Screenshots of Sheet 2: VLOOKUP-Based Student Result	7
Figure 4: Marksheet of roll no. 1	8
Figure 5: Marksheet of roll no. 2	9
Figure 6: Marksheet of roll no. 3	10
Figure 7: Marksheet of roll no. 4	11
Figure 8: Marksheet of roll no. 5	12
Figure 9: Screenshots of Percentage	13
Figure 10: Screenshots of Remark	14
Figure 11: Screenshots of Grade	15
Figure 12: Screenshots of Rank	16

Introduction to MS Excel

Microsoft is a spreadsheet program developed by Microsoft. It runs on Windows, macOS, Android and iOS. The first version was released in 1985 and has gone through several changes over the years.

MS Excel is an application program that allows us to create spreadsheet, which are represented in the form of a table containing rows and columns. The horizontal sequence in which the data is stored is referred to as a row. The vertical sequence in which the data is stored is referred to as a column. In a spreadsheet, a row is identified by a row header and a column is identified by a column header. Each value in a spreadsheet is stored in a cell, which is the intersection of rows and columns. A cell can contain either numeric value or a character string. We can also specify the contents of a cell using formulas. In a spreadsheet, we can perform various mathematical operations using formula, such as addition, subtraction, multiplication, division, average and percentage, etc. Excel organizes data in columns and rows and allows you to do mathematical functions.

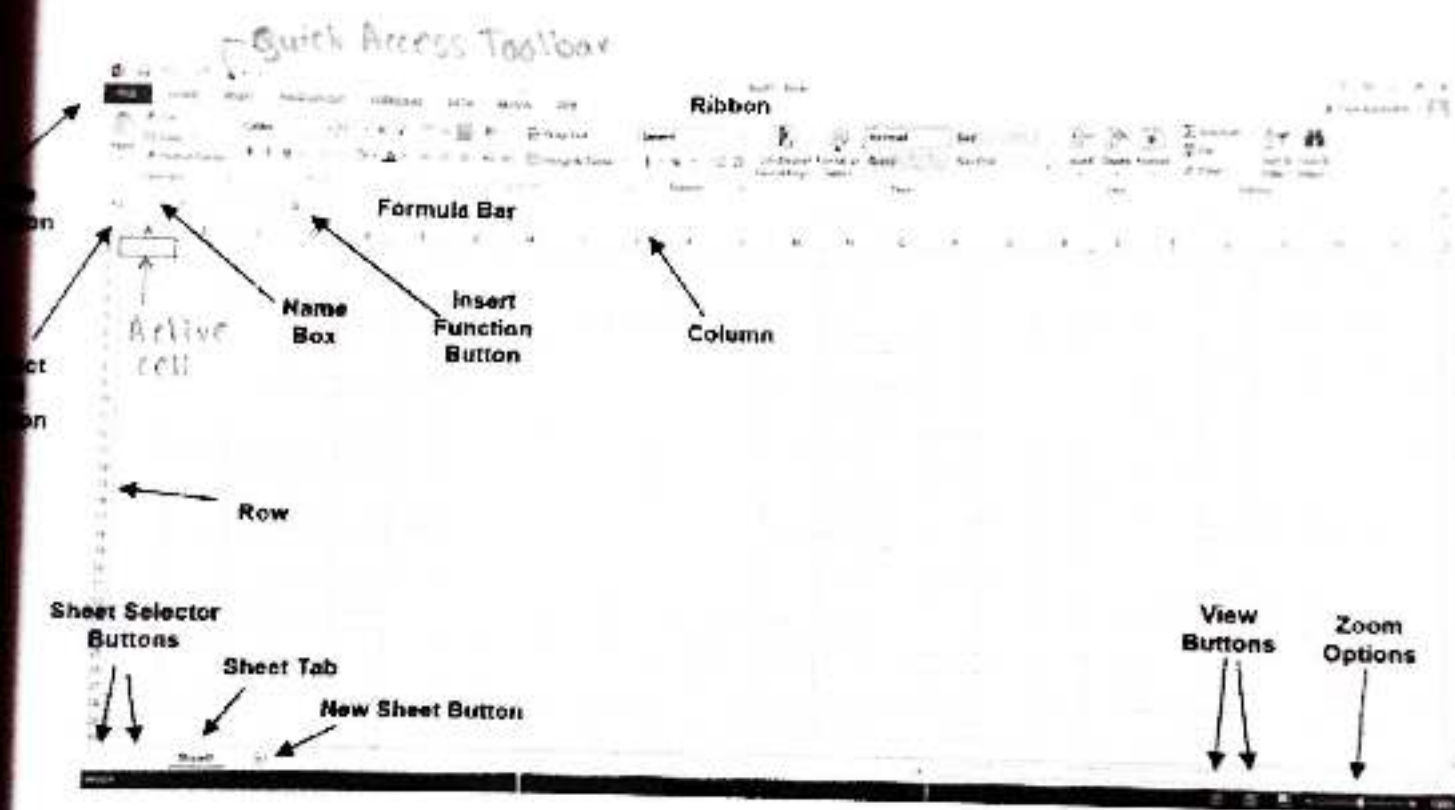


Figure 1: Structural Blueprint of the Microsoft Word User Interface

Cell in MS Excel

A cell is the basic unit of a worksheet in Microsoft Excel. It is formed by the intersection of a row and a column and is identified by a unique cell address. A cell stores different types of data, including text, dates, numbers, formulas and functions. Cells can be edited and formatted to organize and analyze data efficiently. They can also be copied, moved or referenced in formulas to perform various calculations automatically.

Relative Cell Reference

A relative cell reference is the default type of cell reference in Microsoft Excel. It changes automatically when a formula is copied or moved to another cell. Relative references do not use the dollar sign, making them useful for performing the same calculation on multiple rows or columns.

Example: if the formula $= A2 + B2$ is copied from cell C2 to C3, it automatically changes to $= A3 + B3$

Mixed Cell Reference

A mixed cell reference is a combination of relative and absolute cell references. In a mixed reference, either the row or the column is fixed using the dollar sign, while the other changes when the formula is copied. It is useful when one part of the reference must remain constant and the other should change automatically. Mixed references are written as \$A1 or A\$1.

Examples: \$A1 - Column A is fixed, but the row changes
A\$1 - Row 1 is fixed, but the column changes.

Absolute Cell Reference

An absolute cell reference is a cell reference that remains fixed when a formula is copied or moved to another cell. It uses the dollar sign before both the column letter and the row number to lock the cell reference. Absolute references are useful when a formula always needs to refer to the same cell, such as a fixed tax rate or constant value. Using it helps maintain accuracy in calculations by preventing the reference from changing automatically.

VLOOKUP

VLOOKUP is a built in function in Microsoft Excel that is used to search for a specific value in the first column of a table and return the corresponding value from another column in the same row. It helps user quickly retrieve information from large datasets, making data management easier and more efficient. VLOOKUP searches vertically from top to bottom and is commonly used in mark sheets, employee records and databases.

Syntax of VLOOKUP

=VLOOKUP (lookup_value, table_array, col_index_num, [range_lookup])

Example: =VLOOKUP (C13, Sheet1! A1: S11, 10, 0)

- lookup_value (C13) - The value stored in cell C13 that Excel searches for in the first column of the table.
- table_array (Sheet1! A1: S11) - The range of cells on Sheet1 where Excel searches for the lookup value and retrieves the required data.
- col_index_num (10) - Specifies that Excel should return the value from the 10th column of the selected table.
- range_lookup (0) - If indicates an exact match. It is the same as FALSE. If the exact value is not found, Excel returns #N/A.

Reflection

This project helped me understand the practical use of Microsoft Excel in organizing and managing data. While creating the marksheet, I learned how to use formulas, cell references, data validation, and the VLOOKUP function to retrieve information automatically from another worksheet. I also improved my skills in formatting worksheets and handling data accurately. Overall, this project increased my confidence in using Excel for academic and real life tasks. It also improved my problem solving skills while correcting errors and testing formulas. I now have a better understanding of how Excel can simplify complex tasks and improve productivity.

Conclusion

In conclusion, Microsoft Excel is a powerful tool for managing and analyzing data efficiently. By completing this marksheet project, I learned how to create an organized worksheet, apply formulas and functions, and automate data retrieval using VLOOKUP. This project enhanced my understanding of Excel features and demonstrated how they can simplify calculations and reduce manual work. The knowledge and skills gained from this project will be useful for future academic and professional tasks. It also encouraged me to explore more advanced features of Excel for effective data management.